Water Level Monitoring & Alert System

A Mini Project Report Submitted
In partial fulfillment of the requirements for the award of the degree of

Bachelor of Engineering in
Information Technology

by

Tanuja Chavan Roll No 12

Under the esteemed guidance of

Prof. Sonal Patil

Department of Information Technology

Kai. Bhaskarrao Mahajan Samajik Shaikshnik Va Sanskrutik Sanstha's

Indala College of Engineering

(APPROVED BY AICTE DTE, GOVERNMENT OF MAHARASHTRA AND AFFILIATED TO UNIVERSITY OF MUMBAI)

Bapsai, Kalyan, Maharashtra 421103
website: https://icoe.ac.in/

2025-2026

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CERTIFICATE

This is to certify that this is the bonafide record of the Mini Project entitled **"Water Level Monitoring And Alert System"**, submitted by **Tanuja Chavan(12), Harsh Bhosale(08) Sakshi Bombale(09) and Omkar Chaudhari(11)** of TE. in the partial fulfillment of the requirements for the degree of Bachelor in Computer Engineering during the year 2025-2026. The results embodied in this mini project report have not been submitted to any other university or institute for the award of any degree or diploma.

Internal Guide
Prof. Sonal Patil

Head of the Department
Prof. Vrushali Bhamare

External Examiner

<u>DECLARATION</u>

We hereby declare that the mini project titled **“Water Level Monitoring And Alert System”** submitted to **Indala College of Engineering** (Approved by AICTE DTE, government of Maharashtra and affiliated to university of Mumbai) for the award of the degree Bachelor in Information Technology is a result of original work carried-out in this project.It is further declared that the mini project report or any part there of has not been previously submitted to any University or Institute for the award of degree or diploma.

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With regards and gratitude

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ABSTRACT

Water are among the most destructive natural disasters, causing loss of life, damage to infrastructure, and severe economic impact. The Water level Monitoring and Alert System is designed to provide real-time detection, analysis, and early warning of potential water events.

The system integrates water level sensors, rainfall gauges, and IoT-based communication modules deployed in rivers, dams, and Water-prone areas. Sensor data is continuously transmitted to a central server or cloud platform where it is processed using threshold algorithms and predictive models. When water levels exceed predefined safe limits or rapid changes are detected, the system automatically triggers alerts via SMS, mobile app notifications, sirens, and local authority dashboards.

The objective is to reduce response time, enable timely evacuation, and minimize damage by providing accurate, location-specific warnings to residents and disaster management agencies. The system is low-cost, scalable, and can be integrated with weather forecasting data for improved prediction accuracy.

The processing layer employs threshold-based algorithms and machine learning models to analyze incoming data in real time. Historical flood data, current sensor readings, and meteorological API data are combined to improve prediction accuracy and reduce false alarms. When water levels cross predefined warning or danger thresholds, or when the rate of increase indicates flash flood conditions, the system autonomously generates alert triggers.

The alerting layer disseminates warnings through multiple redundant channels to maximize reach: SMS and IVR calls to registered residents, push notifications via a mobile application, activation of local sirens/public address systems, and real-time dashboard updates for disaster management authorities. The dashboard provides geospatial visualization of affected areas, predicted inundation maps, and evacuation route suggestions to support coordinated response.Key advantages of the system include low cost, scalability, minimal human intervention, and 24/7 autonomous operation. The modular design allows integration of additional sensors or weather forecasting modules as needed. Pilot implementations show that early warnings of 2–6 hours can be achieved for riverine floods, significantly improving evacuation time and resource mobilization.

This system aims to shift flood management from reactive to proactive by providing timely, accurate, and location-specific information to both the public and authorities, thereby minimizing casualties and economic losses caused by flood events.

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CHAPTER 1

Introduction

Water level monitoring plays a crucial role in managing water resources and preventing damage to life and property. Sudden rise in water levels in reservoirs, tanks, rivers, and drainage systems can lead to overflow, infrastructure damage, and unsafe conditions if not detected in time. Therefore, an efficient and real-time monitoring system is necessary to ensure safety and proper water management.

According to global observations, poor water level management contributes significantly to economic losses and disruption of daily life. In countries like India, rapid urbanization, improper drainage systems, and irregular water flow often create situations where water levels rise unexpectedly. In such conditions, timely monitoring and alerting become essential to prevent damage and ensure quick response.

Traditional water monitoring systems mainly depend on manual observation and periodic checking. These methods are time-consuming, less accurate, and lack real-time responsiveness. Delays in detecting critical water levels can result in overflow, wastage of water, or even hazardous conditions in residential, industrial, and agricultural areas.

To overcome these limitations, this project presents a **Water Level Monitoring & Alert System** based on the **Arduino Uno**. The system continuously monitors the water level using sensors such as the **HC-SR04 Ultrasonic Sensor** along with a water level sensor module for improved accuracy.

When the water level exceeds a predefined safe limit, the system immediately provides alerts using a buzzer and LED indicators. In addition to local alerts, the system uses a **SIM800L GSM Module** to send SMS or call notifications to users, ensuring that alerts are received even without internet connectivity.

The system is powered by a battery, making it suitable for operation during power outages. It is compact, cost-effective, and easy to deploy in various environments such as homes, industries, water tanks, and drainage systems.

Thus, this project demonstrates an effective application of embedded systems for real-time water level monitoring and alerting, helping to improve safety, reduce water wastage, and enable timely response to critical situations.

1.1 Problem Definition:

Water are among the most frequent natural disasters in India and many other parts of the world. Every year during the monsoon season, sudden rise in water levels of rivers, dams, and local streams causes severe loss of human life, livestock, crops, and infrastructure. According to disaster management data, a major reason for high casualties is not the flood itself, but the lack of timely and location-specific warnings reaching the people who live in vulnerable areas.

Currently, most monitoring methods depend on manual inspection by local authorities or data from large government-run river gauge stations. These stations are limited in number and are usually installed only on major rivers. Small rivers, local drains, and low-lying urban areas remain completely unmonitored. Even where monitoring exists, the process of data collection, analysis, and dissemination of alerts is slow and involves multiple human steps, which causes critical delay. By the time an official warning is issued through television, radio, or newspapers, water has often already entered residential areas.

Another major issue is communication breakdown during heavy rainfall. Power outages are common, mobile networks get congested, and internet connectivity is unreliable in rural or semi-urban flood-prone zones. Therefore, any alert system that depends only on Wi-Fi or continuous electricity fails exactly when it is needed most.

In addition, existing commercial flood warning solutions are expensive, require complex installation, and need continuous maintenance, making them unaffordable for small villages, schools, or local community use.

Hence, there is a clear need for a compact, low-cost, battery-powered system that can work autonomously 24/7. The system should be able to:

1. Continuously monitor water level using non-contact and contact-based sensors.
2. Detect rainfall as an early indicator of possible flooding.
3. Process sensor data locally using a microcontroller without depending on external servers.
4. Provide immediate local alerts through buzzer and LED indicators for nearby residents.
5. Operate on battery backup during power failures, which are very common in flood situations.

To address these challenges, this project develops a Water level Monitoring and Alert System using Android UNO as the main controller, HC-SR04 Ultrasonic Sensor and Water Level Sensor for level detection, Rain Sensor LM393 for early rainfall sensing, SIM800L GSM Module for remote SMS/call alerts, Active Buzzer and LEDs for local alerts, and a 3.7V Li-ion battery for uninterrupted operation. The system is first prototyped on a breadboard and then made permanent on perfboard using soldering, making it a practical and deployable solution for real-world flood-prone areas.

This system aims to bridge the gap between large-scale government monitoring and ground-level community safety by providing affordable, real-time, and reliable flood alerts.

1.2 Existing System:

The proposed system is a low-cost **Water Level Monitoring and Alert System** designed to provide real-time water level monitoring and multi-channel alerting. The system works autonomously and remains operational during power outages, making it suitable for deployment in both rural and urban areas.

The proposed system is a smart and automatic Water Level Monitoring and Alert System designed to continuously monitor the water level and provide early warning alerts. This system uses an **Arduino Uno**, an **HC-SR04 Ultrasonic Sensor**, a water level sensor, LEDs, a buzzer, and a **SIM800L GSM Module** to detect rising water levels and notify users in real time.

In this system, the ultrasonic sensor is placed above the water surface to measure the distance between the sensor and the water level. As the water level increases, the measured distance decreases. The **Arduino Uno** continuously reads this data and compares it with predefined threshold values.

When the water level reaches a warning level, the system turns on a yellow LED. When the water level becomes dangerous, a red LED and buzzer are activated to alert nearby people.

Additionally, the **SIM800L GSM Module** is used to send SMS alerts to registered mobile numbers. This enables remote monitoring and ensures that alerts are received even if no one is physically present near the system. The water level sensor can also act as an additional safety component for direct water detection.

The system is powered using a 3.7V Li-ion battery, making it portable and suitable for remote locations. The entire setup works automatically without human intervention and provides real-time monitoring, early warning, and improved safety in water level management applications.

1.3 Proposed System:

The proposed Water Level Monitoring and Alert System is designed to automatically monitor the water level and provide early warning alerts to prevent damage caused by rising water levels. The system uses sensors, a microcontroller, and communication modules to detect changes in water level and notify users in real time. This helps people take preventive action before the situation becomes dangerous.

In this system, an **HC-SR04 Ultrasonic Sensor** is used to measure the water level. The sensor is placed above the water surface and continuously measures the distance between the sensor and the water. As the water level rises, the measured distance decreases. This data is sent to the **Arduino Uno**, which processes the readings and compares them with predefined threshold values such as safe level, warning level, and danger level.

When the water level reaches the warning level, the system activates an LED indicator to show that the water level is increasing. If the water level continues to rise and reaches the danger level, another LED turns on and an active buzzer is activated. This provides an immediate alert to nearby people so they can take necessary precautions such as protecting equipment or moving to safer areas.

The system also includes a **SIM800L GSM Module**, which is used to send SMS alerts to registered mobile numbers. When the water level crosses the dangerous threshold, the system automatically sends a warning message such as “Water Level High” to users. This ensures that alerts are received even when no one is physically present near the system.

Additionally, a water level sensor is used as a secondary detection method. This sensor detects the presence of water directly and acts as a backup safety feature. If water reaches the sensor, the system immediately triggers the buzzer and sends an alert message, increasing the overall reliability of the system.

The entire system is powered by a 3.7V Li-ion battery, making it portable and suitable for remote or outdoor locations where power supply may not be available. The system operates automatically without requiring human intervention and provides continuous monitoring, early warning, and improved safety.

Overall, the proposed system reduces manual effort, improves response time, and provides a reliable solution for water level monitoring and alerting in various applications such as water tanks, reservoirs, and drainage systems.

1.4 Literature Review:

Water level monitoring and early warning systems have been an active area of research due to the increasing frequency of flood events caused by climate change and rapid urbanization. Several researchers and organizations have proposed different approaches using sensors, microcontrollers, and wireless communication to detect and communicate flood risks.

1. Traditional Flood Monitoring Methods

Earlier water warning systems were mainly dependent on manual observation of river gauge stations and meteorological data collected by government agencies. According to the Central Water Commission (CWC), India, flood forecasting is done using hydrological models based on rainfall and river discharge data. However, these systems cover only major rivers and have a significant time lag between data collection and public dissemination. The warnings are usually broadcast through TV, radio, or newspapers, which are not effective for immediate local action.

2. Sensor-Based Monitoring Systems

With advancements in embedded systems, researchers started using ultrasonic sensors for non-contact water level measurement. Prachi et al. (2018) developed a river water level monitoring system using HC-SR04 and Arduino, where rising water levels were displayed on an LCD and alerts were sent via GSM. The study showed that ultrasonic sensors provide accuracy of ±1 cm and are suitable for outdoor use. However, the system lacked battery backup and rain detection. Similarly, Kumar et al. (2019) proposed a flood detection system using water level probes and NodeMCU ESP8266. The data was uploaded to the cloud using Wi-Fi, and users could monitor levels through a mobile app. The limitation was complete dependency on internet connectivity, which fails during heavy rainfall and power cuts.

3. Use of GSM for Remote Alerting

To overcome internet dependency, many projects integrated GSM modules like SIM800L or SIM900A. Sharma et al. (2020) designed a flood alert system using Arduino, water level sensors, and SIM800L. The system successfully sent SMS alerts to multiple users when threshold levels were crossed. The study highlighted that GSM-based alerts are more reliable in rural areas where 2G network coverage exists but internet is poor. However, power management was not addressed, and the system stopped working during power failure.

4. IoT and Cloud Integration

Recent works have focused on IoT platforms such as ThingSpeak, Blynk, and Firebase for real-time data logging. Singh and Patel (2021) implemented a flood monitoring system using ESP8266, HC-SR04, and ThingSpeak cloud. The advantage was graphical visualization and historical data analysis. The drawback was that cloud dependency made it unsuitable for areas with no internet during disasters.

5. Inclusion of Rainfall Detection

Some researchers integrated rain sensors to improve early prediction. Nair et al.

(2022) used LM393 rain sensor along with water level sensors to create a two-stage alert system. If heavy rain was detected continuously for 30 minutes, a pre-warning was issued even before water level rose.

This reduced false alarms and increased preparedness time. The system still lacked local audio-visual alerts for on-site people.

6. Battery-Operated and Low-Cost Systems

Studies by Rathod et al. (2021) emphasized the need for battery-operated systems using Li-ion cells and solar charging for uninterrupted operation. Their prototype used ESP32, buzzer, and LEDs for local alerts, proving that low-cost components can build effective community-level warning systems. The research concluded that combining ultrasonic + contact-type water sensors increases reliability because ultrasonic readings can be affected by floating debris.

CHAPTER 2

2.1 Hardware & software requirement

Hardware requirement:

Hardware Requirements
The hardware components used in this project are listed below. These components help in detecting water level, processing data, and giving alerts.

* Android UNO – Main microcontroller
* HC-SR04 Ultrasonic Sensor – Water level distance measurement
* Water Level Sensor – Detect water presence
* SIM800L GSM Module – Send SMS alerts
* Active Buzzer (5V) – Audio alert
* LED (2 or more, multi-colour) – Visual indication
* 220 Ohm Resistors – For LED protection
* Momentary Tactile Push Button (5V) – Reset/manual control
* Li-ion Battery (3.7V) – Power supply
* Breadboard (830 point) – Circuit testing
* Perfboard (7×9 cm or 9×18 cm) – Permanent circuit mounting
* Jumper Wires (MM, MF, FF) – Connections
* Solid Core Wire – Permanent wiring
* Soldering Iron, Solder Wire, Flux (Wax) – Soldering connections

Software requirement:

Software Requirements —
The software tools used to develop and program the system are:

* Arduino IDE – Writing and uploading code
* Arduino Uno
* Embedded C / Arduino Programming Language – Code development
* Serial Monitor – Debugging and testing
* GSM AT Commands – For SIM800L message sending
* USB Driver (CP2102 / CH340) – Arduino Uno connection to computer

These hardware and software requirements together help to build and operate the Smart Water Level Monitoring and Early Warning System successfully.

2.1 Software Requirement Specification (SRS):

1.1 Purpose

The purpose of this Water level Monitoring and Alert System is to monitor water level and rainfall conditions. The system detects flood situations and sends alerts using buzzer, LED indicators, and SMS notifications to warn users in advance.

1.2 Scope

The system will:

- Monitor water level using ultrasonic sensor
- Detect rainfall using rain sensor
- Identify flood conditions
- Activate buzzer alert
- Display LED indication
- Send SMS alert using SIM800L
- Provide manual reset button

2. Overall Description

2.1 System Overview

The system reads data from sensors and compares it with predefined threshold values. When the water level increases beyond the safe limit or rainfall is detected, the system triggers alerts.

2.2 System Users

- Local residents
- Disaster management authorities
- River monitoring authorities
- Rural flood monitoring systems

2.3 Operating Environment

- Android UNO
- Arduino IDE
- Embedded C / Arduino programming
- SIM800L GSM module
- Battery powered system

3. Functional Requirements

3.1 Water Level Monitoring

- The system shall measure water level using HC-SR04 ultrasonic sensor

- The system shall continuously monitor water level
- The system shall calculate distance and determine water level

3.2 Rain Detection

- The system shall detect rain using LM393 rain sensor
- The system shall activate warning mode when rain is detected

3.3 Water Detection

The system shall define three levels:

- Safe Level
- Warning Level
- Danger Level

3.4 Alert System

When danger level is reached:

- Buzzer shall turn ON
- Red LED shall turn ON
- SMS alert shall be sent

3.5 SMS Notification

- The system shall send SMS using SIM800L GSM module
- The system shall send warning and danger alerts

Example Message:

"Flood Alert! Water level is high. Take precautions."

3.6 LED Indication

- Green LED → Safe Level
- Red LED → Danger Level

3.7 Manual Reset

- The system shall reset alerts when push button is pressed
- The buzzer shall turn OFF after reset

4. Non-Functional Requirements

4.1 Performance Requirement

- The system shall operate in real-time
- Alert delay shall be less than 5 seconds

4.2 Reliability

- The system shall operate continuously
- Sensor readings shall be accurate

4.3 Power Requirement

- The system shall operate on 3.7V Li-ion battery
- The system shall consume low power

4.4 Safety Requirement

- The system shall detect overflow conditions

- The system shall provide early warning

4.5 Accuracy Requirement

- Water level measurement shall be accurate
- Rain detection shall be reliable

5. Software Requirements

5.1 Development Software

- Arduino IDE

5.2 Programming Language

- Embedded C
- Arduino C++

5.3 Libraries Required

- **NewPing.h** – for ultrasonic sensor (**HC-SR04 Ultrasonic Sensor**)
- **SoftwareSerial.h** – for GSM communication (**SIM800L GSM Module**)
- **Wire.h** – for I2C communication (optional, if needed)

5.4 Drivers Required

- CH340 / CP2102 USB driver

6. Input Requirements

System inputs include:

- Ultrasonic sensor input
- Water level sensor input
- Push button input

7. Output Requirements System outputs include:

- Buzzer alert
- LED indication
- SMS notification
- Serial monitor output

8. System Constraints

- GSM network must be available
- Battery backup is limited
- Sensor range is limited
- Weather conditions may affect accuracy

9. Assumptions

- Sensors are properly installed
- GSM signal is available
- Battery is charged

CHAPTER 3

System design

1. System Architecture:
The system architecture of the Water level monitoring and alert system
consists of input sensors, a processing unit, alert devices, and communication module. All components work together to monitor water level and provide early warning alerts.

1.1 Input Layer (Sensors)
The input layer includes sensors that collect real-time environmental data:
- HC-SR04 Ultrasonic Sensor – Measures distance of water level
- Water Level Sensor – Detects presence of water
- Push Button – Manual reset input

These sensors send signals to the Arduino Uno.

2.2 Processing Layer (Controller)
The Android UNO acts as the main controller. It performs the following
tasks:
- Reads data from sensors
- Processes water level values
- Compares values with threshold levels
- Decides safe, warning, or danger condition
- Controls output devices
- Sends command to GSM module

3.3 Output Layer (Alert System)
The output layer provides alerts to users:
- LED 1 – Safe/Warning level indication
- LED 2 – Danger level indication
- Active Buzzer – Audio alert for high water level

4.4 Communication Layer
- SIM800L GSM Module – Sends SMS alert to users when water level crosses danger level.

5.5 Power Supply
- 3.7V Li-ion Battery supplies power to the entire system.

System Working Flow
Sensors → Android UNO → Decision Making → LED/Buzzer Alert → GSM SMS Notification → User
This architecture allows real-time monitoring, early warning alerts, and remote

notification for flood prevention.

UML Diagram

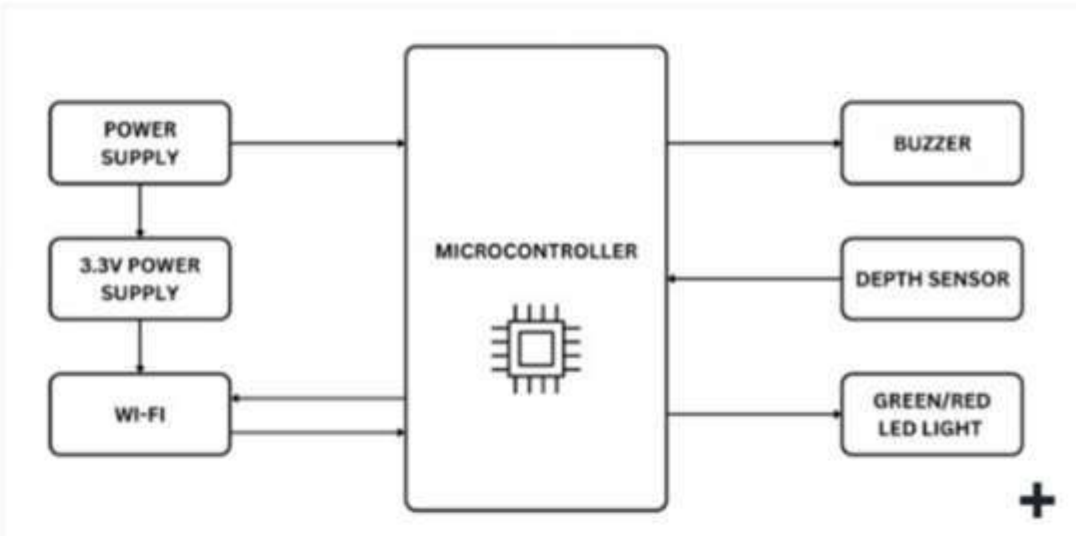


Fig: Water level monitoring and alert system

CHAPTER 4

Implementation

The Water Level Monitoring and Alert System is implemented to monitor water level and rainfall conditions in real time. The system uses Android UNO as the main controller along with ultrasonic sensor, water level sensor, rain sensor, SIM800L GSM module, buzzer, and LED indicators. The main objective of this implementation is to detect flood conditions early and provide warning alerts.

In this implementation, the sensors continuously monitor environmental conditions. The ultrasonic sensor measures the distance between the sensor and water surface to determine the water level. The water level sensor detects the presence of water and helps in identifying overflow situations. The rain sensor detects rainfall and provides additional information about weather conditions.

All the sensor data is sent to the Android UNO. The controller processes the received data and compares it with predefined threshold values. These threshold values are set to identify three conditions: safe level, warning level, and danger level. When the water level is below the safe limit, the system remains in normal mode and indicates safe condition.

When the water level starts increasing and reaches the warning level, the system changes to warning mode. In this stage, the system alerts users using LED indication. This warning helps users to take precautions before the situation becomes dangerous.

If the water level continues to rise and reaches the danger level, the system activates the alert mechanism. The buzzer turns ON to provide an audible alert. At the same time, the SIM800L GSM module sends an SMS alert message to the predefined mobile number. This message informs users about the flood condition and warns them to take immediate action.

The rain sensor also plays an important role in this implementation. When heavy rain is detected, the system increases alert sensitivity. This helps in predicting possible flood situations even before the water level becomes dangerous.

A push button is included in the system for manual reset. When the button is pressed, the system stops the buzzer and resets the alert condition. After reset, the system again starts monitoring continuously.

The entire implementation works in real time and provides early flood warning. The system is low cost, easy to install, and suitable for rivers, dams, drainage systems, and flood-prone areas. This implementation helps in reducing flood damage and improves safety by providing timely alerts.

4.1 Sample Code

```
// Water Level Monitoring and Alert System
#include <SoftwareSerial.h>
// GSM pins
SoftwareSerial gsm(2, 3); // RX, TX
// Pins
int trigPin = 9;
int echoPin = 10;
int waterSensorPin = A0;
int buzzerPin = 8;
int redLED = 6;
int greenLED = 5;
int yellowLED = 4;
int buttonPin = 7;
// Variables
long duration;
int distance;
int waterValue;
bool buzzerMute = false;
bool lastButtonState = HIGH;
bool smsSent = false;   // prevent spam
void setup() {
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);
  pinMode(buzzerPin, OUTPUT);
  pinMode(redLED, OUTPUT);
  pinMode(greenLED, OUTPUT);
  pinMode(yellowLED, OUTPUT);
  pinMode(buttonPin, INPUT_PULLUP);
  Serial.begin(9600);
  gsm.begin(9600);
  delay(2000);
  // GSM setup
  gsm.println("AT");
  delay(1000);
  gsm.println("AT+CMGF=1"); // SMS text mode
  delay(1000);
```

```
}
void sendSMS() {
 gsm.println("AT+CMGF=1");
 delay(1000);
gsm.println("AT+CMGS=\"+91 9321679780\"");
 delay(1000);
 gsm.print("ALERT: Water Level High!");
 delay(1000);
 gsm.write(26); // CTRL+Z
 delay(3000);
}
void loop() {
 //  Button → toggle buzzer mute
   bool currentState = digitalRead(buttonPin);

 if (lastButtonState == HIGH && currentState ==
LOW) {
     buzzerMute = !buzzerMute;
    delay(200);
   }
   lastButtonState = currentState;
   //  Ultrasonic
   digitalWrite(trigPin, LOW);
   delayMicroseconds(2);
   digitalWrite(trigPin, HIGH);
   delayMicroseconds(10);
   digitalWrite(trigPin, LOW);
   duration = pulseIn(echoPin, HIGH);
   distance = duration * 0.034 / 2;
   // Water Sensor
   waterValue = analogRead(waterSensorPin);
  Serial.print("Distance: ");
  Serial.print(distance);
   Serial.print(" cm | Water: ");
   Serial.println(waterValue);
 //  MAIN LOGIC
 //  DANGER
```

```
if ((distance > 0 && distance < 10) ||
(waterValue > 400)) {
 digitalWrite(redLED, HIGH);
  digitalWrite(yellowLED, LOW);
  digitalWrite(greenLED, LOW);
//  buzzer
  if (!buzzerMute) {
    digitalWrite(buzzerPin, HIGH);
  } else {
digitalWrite(buzzerPin, LOW);
  }
// Send SMS only once
  if (!smsSent) {
  sendSMS();
  smsSent = true;
  }
}

// SAFE
else {
  digitalWrite(redLED, LOW);
  digitalWrite(yellowLED, LOW);
  digitalWrite(greenLED, HIGH);
  digitalWrite(buzzerPin, LOW);

  buzzerMute = false;
  smsSent = false; // reset for next alert
}

delay(200);
}
```

CHAPTER 5

Results

Testing- Fig 1: Safe Zone

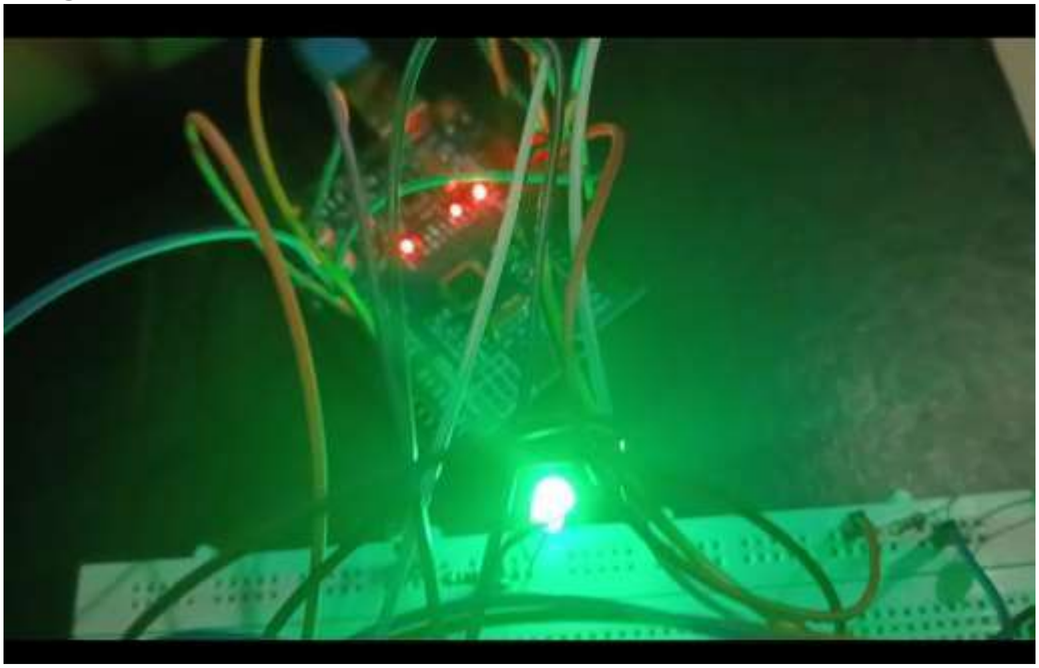
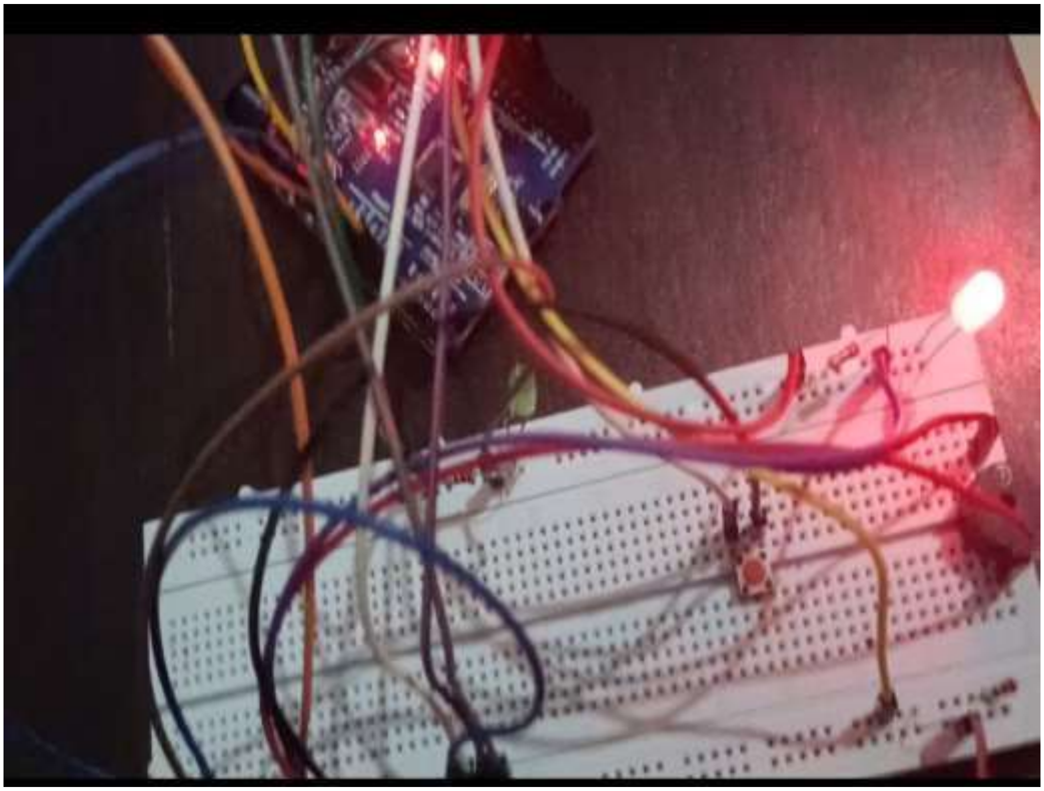


Fig 2: Danger Zone



Output screens: fig 3 – Uploading code

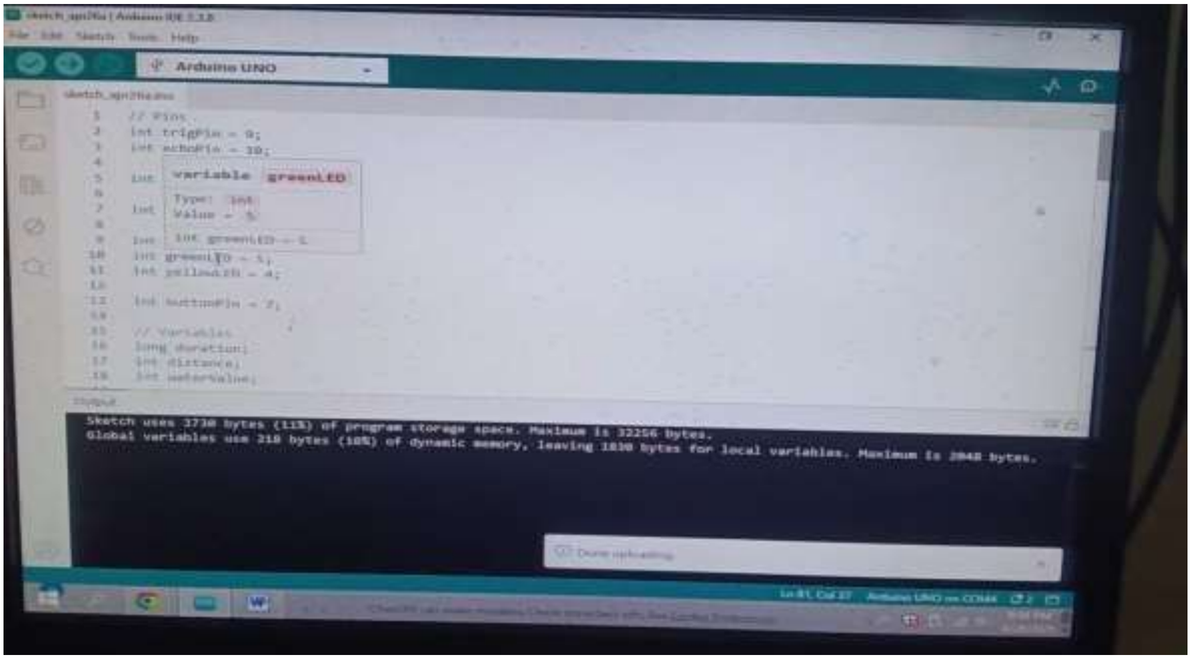
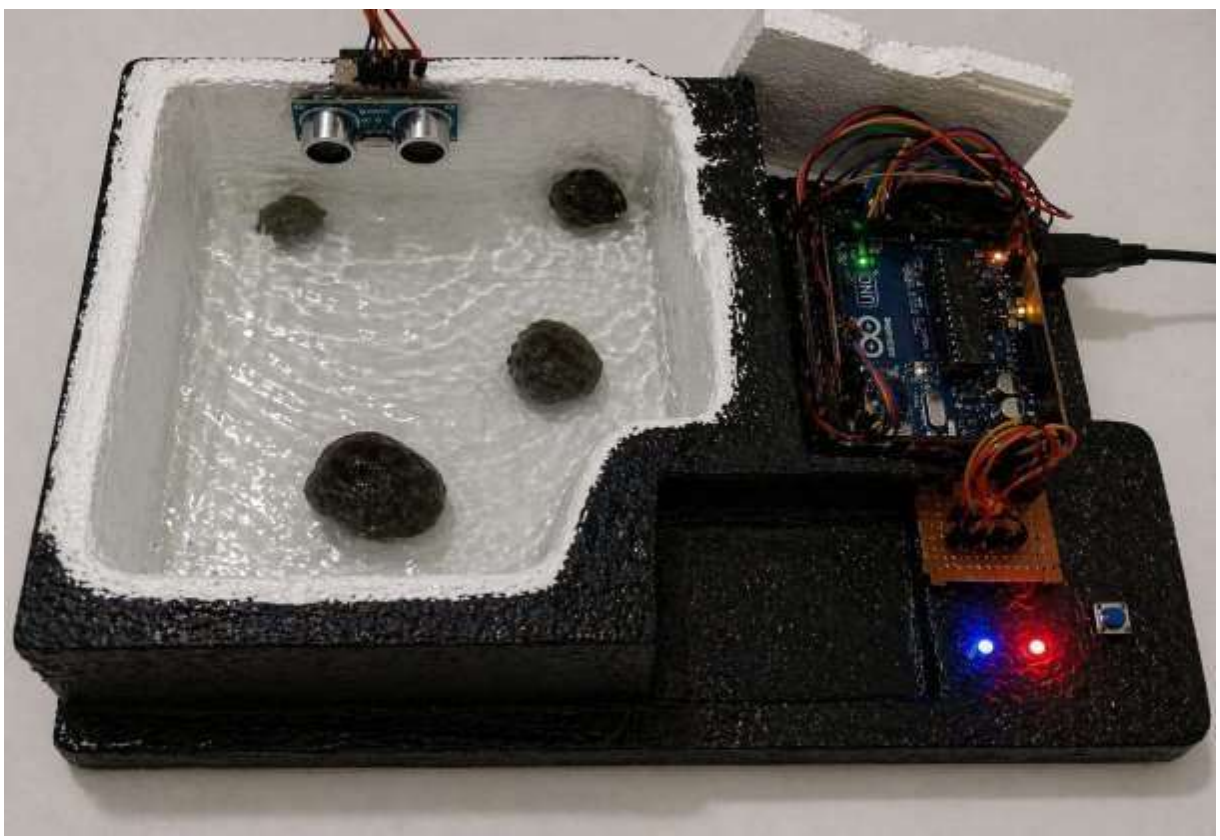


Fig 4: Final Result



Future Scope

1. **IoT Integration (Cloud Monitoring)**
   The system can be upgraded by connecting it to the internet for real-time data monitoring on cloud platforms, allowing users to access water level data remotely.
2. **Mobile Application Development**
   A dedicated mobile application can be developed to display live data, alerts, and system status, making the system more user-friendly and accessible.
3. **Data Logging and Analysis**
   The system can be enhanced with data logging features to store historical water level data, which can be used for analysis and future predictions.
4. **Solar Power Integration**
   Solar panels can be integrated with the system to make it energy-efficient and suitable for long-term outdoor and remote area deployment.
5. **Multi-Point Monitoring System**
   Multiple sensors can be deployed at different locations and connected to a single system to monitor large areas such as reservoirs, dams, or industrial setups.
6. **Automatic Water Control System**
   The system can be extended to automatically control water pumps or valves based on water level conditions, reducing manual intervention.
7. **Advanced Alert Mechanisms**
   In addition to SMS alerts, the system can be upgraded to include voice call alerts, mobile notifications, and siren systems for more effective warning.

CONCLUSION

The Water Level Monitoring and Alert System is developed to monitor water level and rainfall conditions and provide early warning during flood situations. The system uses sensors to continuously collect environmental data and the And processes this data to determine the water level status. Based on predefined threshold values, the system identifies safe, warning, and danger levels.

When the water level is within the safe range, the system continues normal monitoring. When the water level starts increasing, the system enters warning mode and indicates that the situation needs attention. If the water level reaches the danger level or heavy rain is detected, the system activates the alert mechanism. The buzzer turns ON, LED indicators show danger status, and an SMS alert is sent using the SIM800L module. This helps in informing users immediately about possible flood conditions.

The system works in real time and provides continuous monitoring. It is designed using low-cost components, making it affordable and easy to implement. The use of GSM communication allows remote alert notification, which increases the usefulness of the system.

The push button reset feature also helps in controlling the alert manually and restarting the monitoring process.

This project can be used in flood-prone areas such as riversides, dams, bridges, and drainage systems. It helps in reducing risk by providing early warnings and allows people to take preventive actions. The system is simple, reliable, and efficient for flood monitoring.

Overall, the Flood Monitoring and Alert System successfully demonstrates a smart and practical solution for detecting rising water levels and sending timely alerts. This project improves safety, supports disaster management, and helps in minimizing damage caused by floods.

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